1**REVIEW FOR EXAM III Chem 1020**

General things:
1. Study your notes; I have tried to organize the material putting emphasis on areas I believe are most important (and therefore will focus the exam on).
2. Read the two chapters: Chp 15, Chp 16 (names, properties, reactions, etc.)

Chapter 15 - Carboxylic Acids and Esters
1. carboxylic acids are *weak acids*, tending to lose a H^+ to form an acidic solution;
be able to recognize or write the equation showing this property
2. IUPAC nomenclature and some common names (acetic acid (vinegar), formic acid (ants))
3. physical properties, very high boiling points, H-bonds, there are 2 O atoms, including an -OH, sharp smell
4. carboxylic acids will react with bases (especially strong bases like NaOH) to form a salt plus water; this is the neutralization reaction
5. esters (fruity, flowery smell) carboxylic acid part plus alcohol part. Reaction 3, p. 509
6. Nomenclature of esters, carboxylic acid part plus alcohol part
7. Reactions 1-4, 6-7, p. 509; Acid hydrolysis and base hydrolysis of esters

Chapter 16 - Amines and Amides
1. how they're related to the inorganic molecule ammonia (NH_3) ; 1°, 2°, 3° amines
2. nomenclature systems (common and IUPAC) of amines and anilines
3. physical properties–which can H-bond with themselves? which cannot? how does this affect their boiling points? 3° amines cannot H-bond so their boiling points are somewhat lower (for a given #C). Yet, why are they all somewhat equally soluble in water? The H of water can H-bond to the N of any amine. Foul odors of rotting flesh, stinky fish, e.g. cadaverine and putrescine.
4. Most important chemical property is *basicity*. Be able to recognize or write a chemical equation showing an amine acting as a base when dissolved in water (Rxn 1, p. 540)
5. amines are bases, so they react with strong acids; amines react with HCl to form "hydrochloride salts" (Rxn 2, p. 540); the "original free base" can be recovered by reacting the salt with strong base, NaOH (Rxn 3, p. 541)
6. An amide is somewhat like an amine plus a carbonyl, although it cannot be prepared this way. Amides can be prepared by reacting an acid chloride and amine (Rxn 4, p. 541)
7. Amides undergo acid and base hydrolysis like esters, Reaction 6 and 7, p. 541.
8. Amide nomenclature, e.g. N,N-diethylethanamide
9. Amides are neither acids nor bases, they are neutral, but are reasonably water soluble and almost always solids.
10. Amines and Amides are important as biological molecules (read up on *neurotransmitters* and *alkaloids*), in medicines and in illicit drugs.

There will likely be another match the functional group and match the name question. Learn ALL functional groups on p. 360, Table 11.2, backwards and forwards. There is no excuse to miss any of these, and you cannot hope to move forward in the course without it.